

Information

Bayesian Classification

Consider Bayes' Rule for classification, as presented in the lecture slides:

$$P(\omega_i | \mathbf{x}) = \frac{p(\mathbf{x} | \omega_i) P(\omega_i)}{p(\mathbf{x})}$$

Assume we have a classification problem with 5 (fish) classes, a 100-dimensional real-valued feature space, and a training set $\mathcal{D}$ consisting of 5000 examples (fish).

Information

Which of the following statements are correct?

Frage 1

Richtig

Erreichte Punkte 1,00 von 1,00

For any fish $\mathbf{x} \in \mathcal{D}$, $p(\mathbf{x}) \leq 1/5000$

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 2

Richtig

Erreichte Punkte 1,00 von 1,00

For any fish $\mathbf{x} \in \mathcal{D}$, $p(\mathbf{x})$ is a vector with 100 values

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 3

Richtig

Erreichte Punkte 1,00 von 1,00

For any $\mathbf{x}$, $\sum_i p(\mathbf{x} \mid \omega_i) = 1.0$

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 4

Richtig

Erreichte Punkte 1,00 von 1,00

$P(\omega_1 \mid \mathbf{x})$ is a vector with 5 entries

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Information

Consider the (5×5) zero-one loss function (λ) for the above classification problem, a specific set of prior and class-conditional distributions learned from some training set, and a Bayesian Minimum-Risk Classifier as described in the lecture slides.

Which of the following statements are correct?

Frage 5

Richtig

Erreichte Punkte 1,00 von 1,00

When tested on some specific test set $\mathcal{T}$, the minimum-risk classifier will end up in exactly the same position in ROC space as a MAP classifier

- ☒ Wahr ✓
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 6

Richtig

Erreichte Punkte 1,00 von 1,00

$$\sum_i \sum_j \lambda_{ij} = 1.0$$

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 7

Richtig

Erreichte Punkte 1,00 von 1,00

If we multiply all values in λ by 5, the result will be an invalid loss function

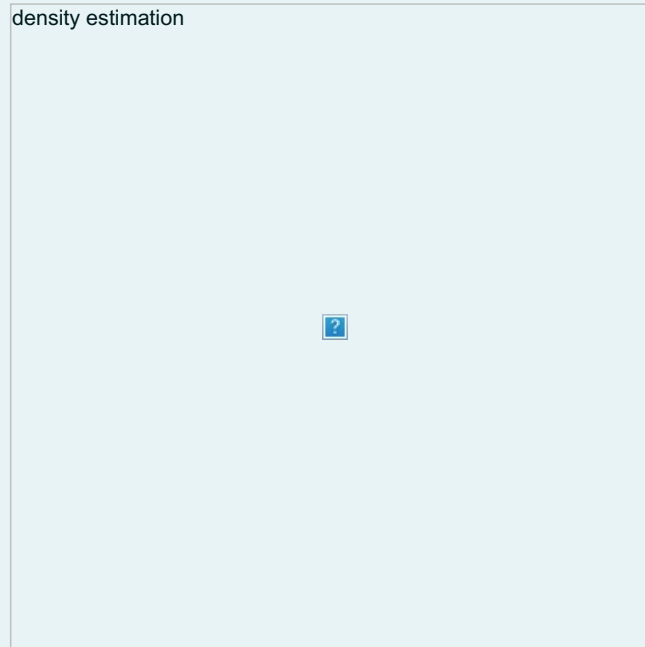
- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Information

Density Estimation

Consider the following figure from the lecture slides, in the context of the task of estimating the mean θ of a normal distribution (with given fixed variance) from a set of observations in a 1D dataset:



Answer the following questions:

Frage 8

Falsch

Erreichte Punkte 0,00 von 1,00

The likelihood function cannot be negative anywhere

- ☐ Wahr
- ☒ Falsch ✖

Die richtige Antwort ist 'Wahr'.

Frage 9

Richtig

Erreichte Punkte 1,00 von 1,00

The y values in the lower plot cannot be probabilities, because they are negative

- ☒ Wahr ✔
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 10

Richtig

Erreichte Punkte 1,00 von 1,00

In the figure, the likelihood function is the plot

- ☐ at the top
- ☐ at the bottom
- ☒ at the middle ✓

Die richtige Antwort ist:
at the middle

Frage 11

Falsch

Erreichte Punkte 0,00 von 1,00

The likelihood function is itself a normal distribution

- ☒ Wahr ✗
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 12

Richtig

Erreichte Punkte 1,00 von 1,00

$\hat{\theta}$ is the maximum probability of the dataset

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 13

Richtig

Erreichte Punkte 1,00 von 1,00

We need to plot the likelihood function in order to find the maximum

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Information

Nearest Neighbour Classifiers

Consider a k-nearest-neighbour classifier in a d-dimensional numeric feature space, with a training set $\mathcal{D}$ containing > 0 instances of two classes, and with the Euclidean distance as distance function.

Which of the following statements (if any) are true?

If we set $k = N$, where $N = |\mathcal{D}|$ is the number of given training examples, then ...

Frage 14

Richtig

Erreichte Punkte 1,00 von 1,00

The classifier will maximally overfit on the training data

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 15

Falsch

Erreichte Punkte 0,00 von 1,00

The resulting classifier will have an expected accuracy of $\sqrt{1/N}$

- ☒ Wahr ✗
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 16

Falsch

Erreichte Punkte 0,00 von 1,00

Each example from the training set will be classified into the correct class

- ☒ Wahr ✗
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 17

Falsch

Erreichte Punkte 0,00 von 1,00

The resubstitution estimate on $\mathcal{D}$ will be 1.0

☒ Wahr ✖

☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 18

Falsch

Erreichte Punkte 0,00 von 1,00

All test examples will be classified into the same class

☐ Wahr

☒ Falsch ✖

Die richtige Antwort ist 'Wahr'.

Information

Trees and Forests

Consider a decision tree learner that operates as follows: given a training set $\mathcal{D}$ and a set of discrete features $\mathcal{F}$, it randomly selects one feature from $\mathcal{F}$, builds a one-level tree by splitting the data according to this feature, and labels the resulting subnodes as leaves with the majority class in each respective node.

Which of the following statements (if any) are true?

Frage 19

Richtig

Erreichte Punkte 1,00 von 1,00

It makes no sense to combine several such trees into a Random Forest, because the trees rely on incompatible features

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 20

Richtig

Erreichte Punkte 1,00 von 1,00

The chosen feature split might have an information gain of zero

- ☒ Wahr ✓
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 21

Richtig

Erreichte Punkte 1,00 von 1,00

Any such decision tree will have an accuracy on the training set $\mathcal{D}$ that is higher than the default accuracy

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 22

Richtig

Erreichte Punkte 1,00 von 1,00

Any such decision tree will have exactly the default accuracy on the training set $\mathcal{D}$

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 23

Richtig

Erreichte Punkte 1,00 von 1,00

The resulting structure is not a decision tree

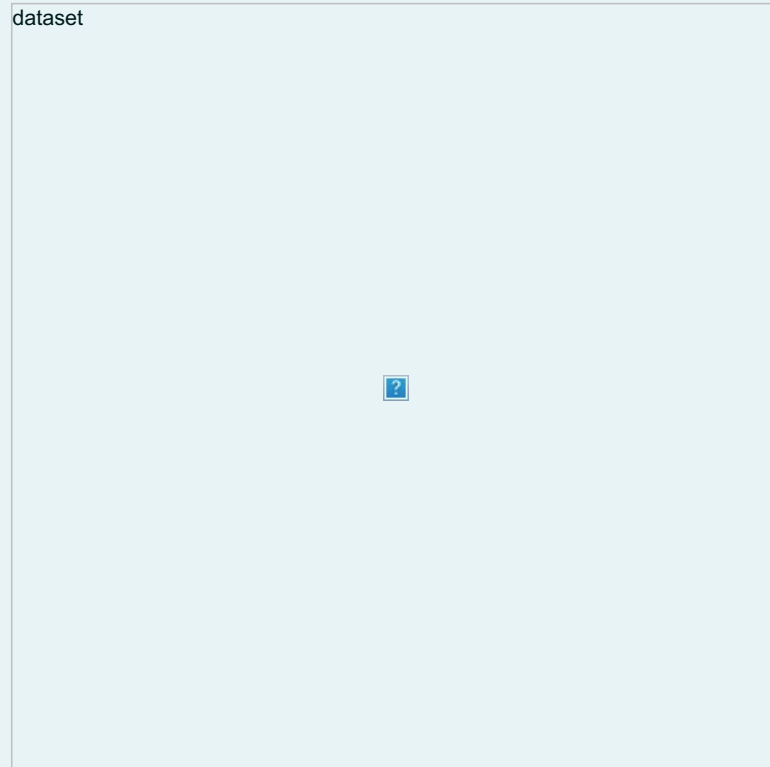
- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Information

Various Classifiers

Consider the following 2-class dataset (from the lecture slides) in a two-dimensional feature space defined by features x_1, x_2 , where the two classes are indicated by filled vs. empty dots:



Information

Which of the following statements are correct?

Frage 24

Richtig

Erreichte Punkte 1,00 von 1,00

A support vector machine with a quadratic kernel can achieve an error of zero on this dataset

- ☒ Wahr ✓
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 25

Falsch

Erreichte Punkte 0,00 von 1,00

If we map this dataset into a new 2D feature space defined by features (x_1^2, x_2^2) , ID3 can separate the two classes with a two-level decision tree (i.e., two levels of splits)

- ☒ Wahr ✖
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 26

Richtig

Erreichte Punkte 1,00 von 1,00

A decision tree learned by ID3 would have to have an infinite number of splits (i.e., infinite depth) in order to achieve an error of zero on this dataset

- ☐ Wahr
- ☒ Falsch ✔

Die richtige Antwort ist 'Falsch'.

Frage 27

Falsch

Erreichte Punkte 0,00 von 1,00

A nearest-neighbour classifier can only perform well on this dataset if it uses circular (and not rectangular) neighbourhoods

- ☒ Wahr ✖
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 28

Richtig

Erreichte Punkte 1,00 von 1,00

The max margin hyperplane that an SVM with quadratic kernel would learn on this dataset remains the same, regardless of which of the two classes we label as +1 and which one as -1

- ☒ Wahr ✔
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 29

Richtig

Erreichte Punkte 1,00 von 1,00

There exists no k for which a k -NN algorithm would predict class "white" for point (0,0)

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Information

Consider a classifier for the above dataset in the form of a Feedforward Neural Network, with two input units for the two features (x_1, x_2) , one hidden unit with the $\tanh$ activation function, and one output unit with identity activation. We represent the two classes in the data as 0 (white) and 1 (black), learn model parameters by training on the dataset with the squared error loss function, and will predict class 1 (black) for an input when the value predicted at the output is higher than some threshold τ , class 0 (white) otherwise. There is no restriction on the size of weights.

Which of the following statements about this model are correct:

Frage 30

Falsch

Erreichte Punkte 0,00 von 1,00

Increasing the classification threshold τ cannot increase the recall of the model on class 0 on the training set

- ☒ Wahr ✗
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 31

Richtig

Erreichte Punkte 1,00 von 1,00

The single hidden unit is unnecessary and makes no difference; a network without hidden unit could learn the same function

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 32

Falsch

Erreichte Punkte 0,00 von 1,00

If we set the threshold τ to 0.0 or 1.0, all possible inputs will always be classified into the same class (black or white, respectively), by any model trained in this way

- ☒ Wahr ✖
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 33

Falsch

Erreichte Punkte 0,00 von 1,00

The output predicted by the output unit is a probability

- ☒ Wahr ✖
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 34

Richtig

Erreichte Punkte 1,00 von 1,00

By choosing different values for τ (for a fixed learned model), we can obtain classifiers with different positions in ROC space

- ☒ Wahr ✔
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 35

Falsch

Erreichte Punkte 0,00 von 1,00

The output predicted by the output unit will always be in the interval $[0, 1]$

- ☒ Wahr ✖
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Information

Hidden Markov Models

Suppose you are (mildly) interested in the currently ongoing European Football Championship. As you go past your neighbour's house, you are trying to infer, from the noises you hear from the open windows, whether there is currently a game going on. You know that when a game is on, your neighbour tends to watch it on TV and, when doing so, will tend to make certain kinds of noises.

Let's model this situation as a HMM with states $S = \{\text{game}, \text{nogame}\}$ and possible noises you might hear $O = \{\text{none}, \text{cheer}, \text{shout}, \text{curse}, \text{snore}\}$, and with parameter values we have somehow learned from observations over many previous championships.

Frage 36

Richtig

Erreichte Punkte 2,00 von 2,00

What is the dimension (number of rows and columns) of matrix $\mathbf{A}$ (as defined in the lecture slides)?

✓ x

✓ (row x col)

Frage 37

Falsch

Erreichte Punkte 0,00 von 2,00

What is the dimension (number of rows and columns) of matrix $\mathbf{B}$ (as defined in the lecture slides)?

✗ x

✗ (row x col)

Frage 38

Falsch

Erreichte Punkte 0,00 von 1,00

What is the dimension of $\mathbf{\Pi}$ (as defined in the lecture slides)?

✗

Information

Which of the following statements about this model are true?

Frage 39

Falsch

Erreichte Punkte 0,00 von 1,00

With only two states, matrix $\mathbf{A}$ must be symmetric

- ☒ Wahr ✖
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 40

Richtig

Erreichte Punkte 1,00 von 1,00

If your neighbour never snores, one row of the matrix $\mathbf{B}$ will be all zeros

- ☒ Wahr ✔
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 41

Falsch

Erreichte Punkte 0,00 von 1,00

Matrix $\mathbf{B}$ represents a Markov Process

- ☒ Wahr ✖
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 42

Richtig

Erreichte Punkte 1,00 von 1,00

The diagonal of $\mathbf{A}$ must sum to 1.0

- ☐ Wahr
- ☒ Falsch ✔

Die richtige Antwort ist 'Falsch'.

Frage 43

Richtig

Erreichte Punkte 1,00 von 1,00

We can estimate the expected duration of a game (= number of consecutive time points $\{t\}$ with $q_t = \text{game}\})$ from

- ☒ matrix $\{\mathbf{A}\}$ of the model ✓
- ☐ vector $\{\mathbf{P}_i\}$ of the model
- ☐ not at all
- ☐ matrix $\{\mathbf{B}\}$ of the model

Die richtige Antwort ist:

matrix $\{\mathbf{A}\}$ of the model